

E-Commerce Data Cleaning & Exploratory Performance Analytics

Transforming raw transactional records into actionable retail intelligence using MySQL Workbench

DOMAIN	DATABASE TOOL	DATASET VOLUME	CORE DELIVERABLE
Global E-Commerce	MySQL Workbench 8.0	500,000+ Transactions	SQL Pipeline & Insights

Executive Summary

An international online retailer generated over 500,000 unorganized transactional records. This study details the end-to-end processing pipeline established in MySQL Workbench to scrub corrupt entries, standardize attributes, and run exploratory queries. The resulting clean dataset uncovered key revenue concentration metrics, geographic growth drivers, and crucial seasonal purchasing spikes.

01 Operational Challenges & SQL Cleaning Protocol

Raw operational logs contained significant noise, missing keys, and invalid transaction entries. Standardizing the dataset required systematic data wrangling queries:

Challenge 1: Unstructured Timestamps Issue: Invoice dates were stored as text strings, preventing time-series analysis. Solution: Applied STR_TO_DATE() to reformat entries into standardized SQL DATETIME records.	Challenge 2: Revenue Distortions Issue: Return invoices ('C' prefix) & \$0.00 promo entries skewed gross figures. Solution: Filtered out cancellations, negative quantities, and zero-value test rows.
Challenge 3: Missing Identifiers Issue: Over 20% of invoice records lacked associated Customer ID tags. Solution: Bifurcated queries into store-wide metrics vs cohort-level buyer profiles.	Challenge 4: Feature Engineering Issue: Lack of pre-computed order revenue and temporal breakdowns. Solution: Created calculated fields (Quantity * UnitPrice) and extracted time features.

02 Geographic Market & Revenue Breakdown

Exploratory queries evaluated transaction volume and gross spend across international operating markets:

MARKET / COUNTRY	TOTAL ORDERS	TOTAL REVENUE (\$)	MARKET SHARE	STRATEGIC STATUS
United Kingdom	~350,000	\$6,854,000	83.8%	Primary Hub
Netherlands	~2,300	\$285,400	3.5%	Expansion Zone
EIRE (Ireland)	~1,800	\$250,200	3.1%	High Growth
Germany	~1,400	\$220,100	2.7%	Stable
France	~1,200	\$197,000	2.4%	Stable

03 Customer Concentration & Purchasing Behavior

Applying Pareto analysis and window functions revealed significant revenue concentration and distinct purchasing patterns across user segments:

Pareto Revenue Distribution

The top **10% of active repeat buyers** generate over **60% of gross store revenue**. Identifying these high-value B2B/wholesale accounts allows for targeted account management and tailored bulk pricing tiers.

Q4 Peak Seasonality

Order volume spikes dramatically from **October through November**, driven by holiday inventory restocking. Preparing warehouse logistics and staffing prior to late Q3 is essential to prevent delivery delays.

Optimal Order Windows

Time-of-day SQL aggregation indicates that order activity peaks between **10:00 AM and 2:00 PM GMT** on weekdays. Marketing teams should schedule promotional announcements and re-engagement emails during these peak active hours.

04 Strategic Recommendations for Business Growth

Based on the quantitative SQL findings, the following actionable strategies are recommended for executive leadership:

1 Diversify Beyond the Primary UK Market

Establish localized European fulfillment partnerships in the Netherlands and EIRE to reduce international shipping friction and capture growing cross-border demand.

2 Automate Reorder Triggers for Top-Tier SKUs

Implement SQL-driven inventory monitoring triggers to ensure the top 20 revenue-generating stock keeping units (SKUs) remain fully stocked leading into the Q4 holiday surge.

3 Develop Dedicated Wholesale Loyalty Tiers

Construct formal incentive programs for top-decile buyers. Offering volume discounts and structured credit terms will lock in high-margin repeat revenue.

4 Standardize Data Capture at Point of Sale

Enforce mandatory customer account creation during online checkout to eliminate missing Customer ID records and improve future customer lifetime value (CLV) tracking.

Technical Verification & Reproducibility

All analytical models, queries, and data transformation procedures documented in this report are fully reproducible via the MySQL script `ecommerce-sales-data-cleaning-eda.sql` hosted in the main project repository.